Module 7 Summative Assessment

Review Learning Objectives

coach

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Assignment details

Due
Dec 8, 11:59 PM IST

Attempts
0 left (1 attempt every 8 hours)

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Submitted
Nov 16, 11:30 PM IST

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Instructions

1. What is the primary goal of maximum likelihood inference? **1 / 1 point**

- [ ] To maximize the posterior probability of the parameters given the data
- [ ] To minimize the variance of parameter estimates
- [ ] To predict future observations
- [x] To find the model parameters that maximize the likelihood of the observed data given the model

Correct
Correct! In maximum likelihood inference, we aim to find the parameters that maximize the likelihood of the observed data given the model.

2. Which of the following situations is NOT suitable for using maximum likelihood estimation? **1 / 1 point**

- [x] Estimating the mean of a sample with many outliers.
- [ ] Estimating the mean of a sample drawn from a normal distribution.
- [ ] Estimating the parameter of an exponential distribution with a large sample size.
- [ ] Estimating the probability of success in a Bernoulli trial.

Correct
Correct. Maximum likelihood estimation is sensitive to outliers, so it might not be the best approach in this case.

3. Which of the following code snippets correctly represents the implementation of a basic Bootstrap procedure in Python? **1 / 1 point**

- [ ] ```python import numpy as np bootstrap_samples = np.random.sample(data, size=(1000, len(data)), replace=True) ```
- [ ] ```python import numpy as np bootstrap_samples = np.random.choice(data, size=(1000, len(data)), replace=False) ```
- [x] ```python import numpy as np bootstrap_samples = np.random.choice(data, size=(1000, len(data)), replace=True) ```
- [ ] ```python import numpy as np bootstrap_samples = np.random.sample(data, size=(1000, len(data)), replace=False) ```

Correct
Correct! This code correctly generates bootstrap samples from the given data.

4. Which of the following is a potential issue when using maximum likelihood estimation for parameters of a statistical model? **0 / 1 point**

- [x] Maximum likelihood estimation always overfits the data.
- [ ] The likelihood function is always a convex function.
- [ ] The likelihood function always has a unique maximum.
- [ ] The likelihood function may have multiple maxima.

Incorrect
This is not necessarily true. Overfitting depends on the complexity of the model and the amount of data, not solely on the estimation method.

5. Why is the Bayesian approach sometimes criticized for its subjectivity? **1 / 1 point**

- [ ] Because the likelihood function is chosen subjectively.
- [x] Because the prior distribution is chosen based on the researcher's subjective beliefs.
- [ ] Because the posterior distribution is derived from subjective data.
- [ ] Because the Bayesian approach requires subjective coding skills.

Correct
Correct! The choice of the prior distribution in Bayesian inference is often based on the researcher's subjective beliefs.

6. Why might Bayesian inference be chosen over Maximum Likelihood Estimation (MLE) in some situations? **1 / 1 point**

- [x] Bayesian inference incorporates prior knowledge, while MLE does not.
- [ ] Bayesian inference always outperforms MLE.
- [ ] Bayesian inference can handle larger datasets than MLE.
- [ ] Bayesian inference is computationally less expensive than MLE.

Correct
Correct! Bayesian inference allows for the incorporation of prior knowledge, which MLE does not.

7. What is the basic principle of Bayesian inference? **1 / 1 point**

- [x] It is a way of updating prior beliefs in light of new data.
- [ ] It is a method to maximize the likelihood of the observed data.
- [ ] It is a technique for estimating population parameters by drawing a sample multiple times.
- [ ] It is a method for making predictions using probability theory.

Correct
Correct! Bayesian inference updates prior beliefs with new data.

8. In maximum likelihood inference, what is the consequence of choosing a model that is too complex for the data? **1 / 1 point**

- [ ] The likelihood will be maximized
- [ ] The model will underfit the data
- [x] The model will overfit the data
- [ ] The likelihood will be minimized

Correct
Correct! A model that is too complex for the data is likely to overfit, fitting the noise in the data as well as the underlying pattern.

9. In maximum likelihood estimation, what is the role of the likelihood function? **0 / 1 point**

- [ ] It specifies the prior probability of the parameters.
- [ ] It specifies the probability of the data given the parameters.
- [x] It specifies the joint probability of the data and the parameters.
- [ ] It specifies the probability of the parameters given the data.

Incorrect
This is incorrect. The likelihood function is not about the joint probability of the data and the parameters.

10. What does the likelihood function represent in the context of Bayesian inference? **1 / 1 point**

- [x] The probability of observing the data given the parameters.
- [ ] The overall probability of observing the data.
- [ ] The chance of the parameters being valid.
- [ ] The probability of the parameters given the data.

Correct
Correct! The likelihood function represents the probability of observing the data given the parameters.

11. Why is bootstrapping often used in conjunction with Bayesian inference? **1 / 1 point**

- [ ] Bootstrapping provides a way to generate additional data for testing.
- [ ] Bootstrapping helps to compute the prior probabilities required for Bayesian inference.
- [ ] Bootstrapping is used to make the Bayesian inference process faster.
- [x] Bootstrapping allows the approximation of the posterior distribution in Bayesian inference.

Correct
Correct! Bootstrapping can help approximate the posterior distribution in Bayesian inference.

12. Which of the following steps would be necessary to create a Bayesian model? **1 / 1 point**

- [x] Defining a likelihood function and a prior distribution.
- [ ] Calculating the maximum likelihood estimate.
- [ ] Determining the correlation coefficient.
- [ ] Performing a t-test.

Correct
Correct! Defining a likelihood function and a prior distribution are essential steps in creating a Bayesian model.

13. In maximum likelihood estimation, why do we often use the log-likelihood instead of the likelihood itself? **1 / 1 point**

- [ ] Because the log-likelihood is always a convex function.
- [ ] Because the maximum of the log-likelihood is different from the maximum of the likelihood.
- [ ] Because the log-likelihood is always larger than the likelihood.
- [x] Because taking the log simplifies the mathematics and numerical stability.

Correct
Correct! Taking the log simplifies multiplication into addition, and it can improve numerical stability when dealing with very small probabilities.

14. What does a high posterior probability indicate in Bayesian inference? **1 / 1 point**

- [ ] The likelihood of the data is very high.
- [ ] The prior beliefs are very accurate.
- [x] The observed data is consistent with the prior beliefs.
- [ ] All observed data are outliers.

Correct
Correct! A high posterior probability indicates that the observed data is consistent with the prior beliefs.

15. Suppose you are using maximum likelihood inference to estimate the parameter of a Poisson distribution. Which of the following is the correct form of the likelihood function? **0 / 1 point**

- [x] L(λ|x) = (e^(-λ) * λ^x) / x!
- [ ] L(λ|x) = λ^x / x!
- [ ] L(λ|x) = (λ^x * e^(-λ)) / x!
- [ ] L(λ|x) = λ^x * e^(-λ)

Incorrect
Remember that the likelihood function for the Poisson distribution includes a factor of x!. Try revisiting the formula.